Prof. Luiz Paulo Lopes Fávero

PRINTS:

$$Y_i = \alpha + \beta_1 X_{1_i} + \beta_2 X_{2_i} + \dots + \beta_k X_{k_i} + \mu_i$$

$$\left.\begin{array}{l} \text{corr} \uparrow [X_1, X_2] \\ \text{corr} \uparrow [X_p, X_q] \end{array}\right\} \underline{\text{Multicolinearidade}}$$

RH

salarios ← econometria

$$\boxed{rh1 = 2 \cdot econ1} \Rightarrow econ1 = \frac{rh1}{2}$$

$$\hat{Salario} = \alpha + \underline{\beta_1} \cdot rh1 + \underline{\beta_2} \cdot econ1$$

$$\hat{salario} = \alpha + \beta_1 \cdot rh1 + \beta_2 \cdot \frac{rh1}{2}$$

$$\hat{salario} = \alpha + \underbrace{\left(\beta_1 + \frac{\beta_2}{2}\right)}_{\beta} rh1$$

$Y = \alpha + \beta_1 X_1 + \beta_2 X_2$

$\text{Tolerance} = 1 - R_p^2$

varia 1 a 0 $\qquad$ 0 – 1

$\dfrac{X_1 = \gamma + \delta X_2}{R_1^2}$

$\text{VIF} = \dfrac{1}{\text{Tolerance}}$

varia 1 a $+\infty$

vif = 10 → Tol = 0,1
$R_p^2 = 0{,}9$

vif = 5 → Tol = 0,2
$R_p^2 = 0{,}8$

Heterorkedasticity

Breusch - Pagan (1980)
Review of Economic Studies.

$u_p = \dfrac{u^2}{\left(\sum_{i=1}^{n} u^2\right)/n} \Rightarrow u_{p_i} = \alpha + \beta . \hat{Y}_i + \varepsilon_i$

Z ANOVA

$\dfrac{SQReg}{2} \rightarrow \chi^2_{1df}$

< 0,05 hetero
> 0,05 homoc.

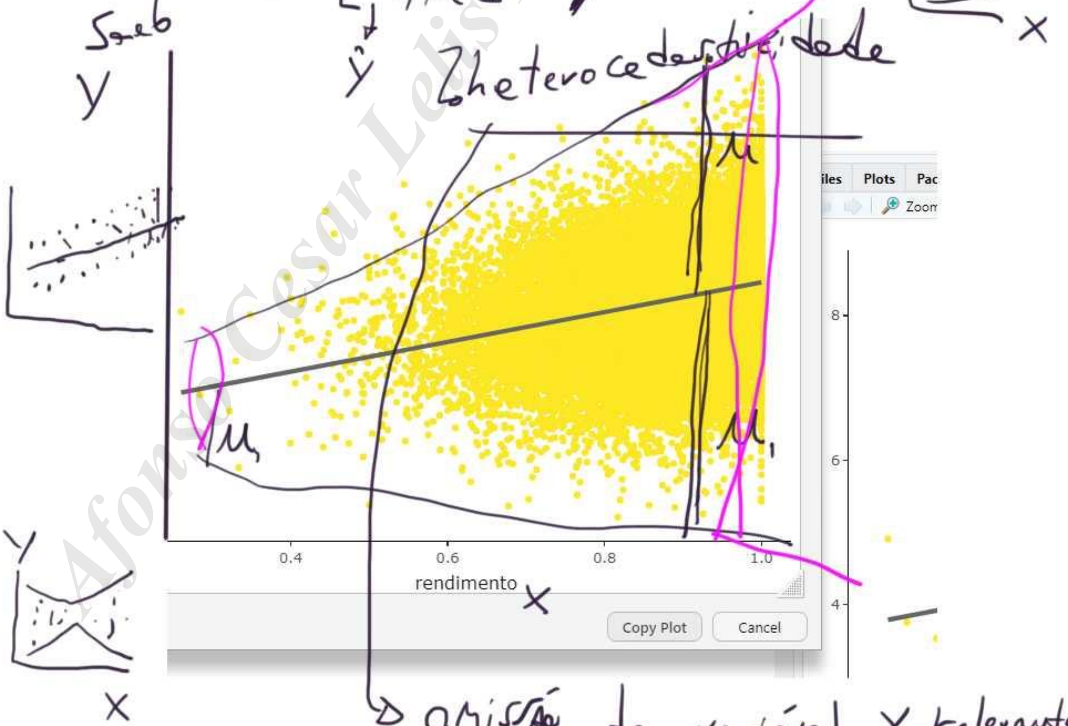
rendimento (fluxo)

saeb

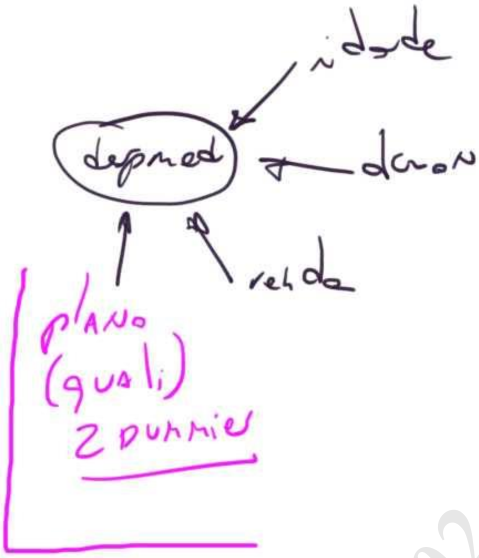
aprendizagem

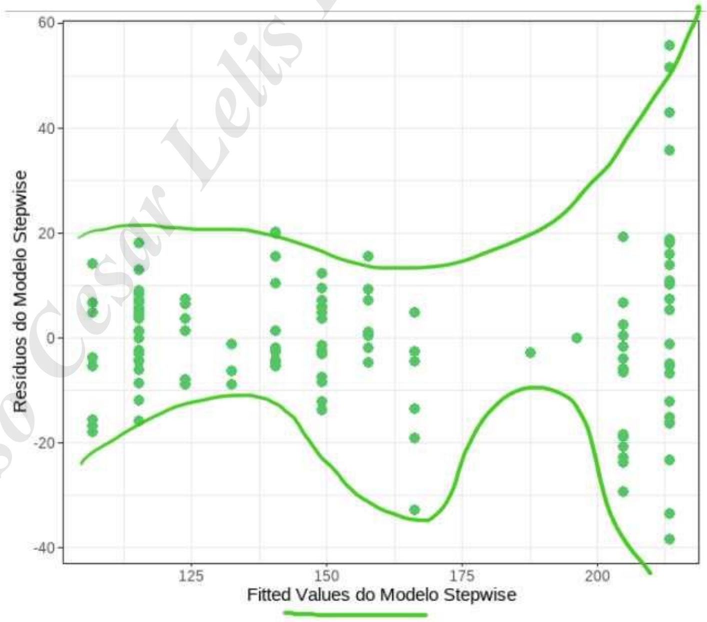
$$\hat{Saeb}_i = \alpha + \beta \cdot rendimento_i$$

$corr[X, M] \approx \emptyset$

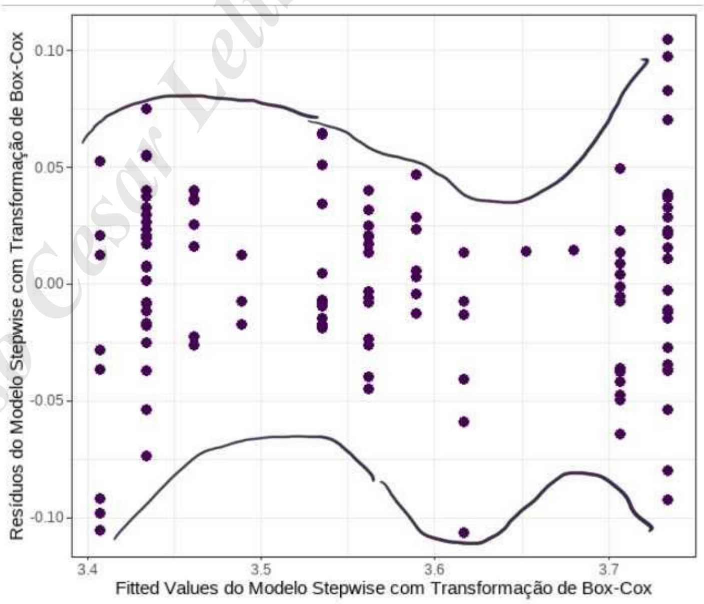


→ omissão de variável X relevantes p/ explicar Y


idade
depmed
dcron
renda
plano
(quali)
2 dummies


Resíduos do Modelo Stepwise
60
40
20
0
-20
-40
125
150
175
200
Fitted Values do Modelo Stepwise

$$\frac{\hat{derpred}_i^{\,-0,144} - 1}{-0,144} = 3,59 + 0,02 \cdot dacon_i - 0,09 \cdot planoesmeralda_i - 0,19\, planoouro_i$$



Resumo (4 aulas OLS):

$$Y_i = \alpha + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + u_i$$

$$\hat{Y}_i = \alpha + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

$$u_i = Y_i - \hat{Y}_i$$

- Dummies (n - 1)
- Diagnósticos:
  - Multicolinearidade (VIF/Toler)
  - Heterocedasticidade (Breusch-Pagan)

- R^2 / $R^2_{aj.}$
- F
- t (α fica!)
- Stepwise
- Box - Cox (Normalidade $\rightarrow$ Shapiro - Francia)
  Erro